PAILA AKASH

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PROFESSIONAL SUMMARY

Third-year Computer Science undergraduate (CGPA: 9.45/10) specializing in Generative AI, full-stack web development, and RF/signal-intelligence systems. Currently interning with the Indian Army's 509 Army Base Workshop (Ministry of Defence) on a counter-UAV detection platform. Co-author of an IEEE conference paper on cognitive digital twins. Proven track record building production-grade full-stack applications, contributing to large open-source codebases, and applying machine learning to real-world defense and healthcare problems.

EDUCATION

K L University, Hyderabad — Hyderabad, India
Bachelor of Technology, Computer Science and Engineering (Gen AI) — **CGPA: 9.45 / 10**

EXPERIENCE

Intern — 509 Army Base Workshop, Indian Army (Ministry of Defence) — Agra, India
Counter-UAV RF Detection & Direction-Finding Project — Jun 2026 – Present

- Contributing to an RF-based UAV detection, direction-finding, and distance-estimation system built around a KrakenSDR coherent receiver, a four-element antenna array, and a Raspberry Pi 5 (8GB), targeting drone signals in the 2.4GHz and 5.8GHz bands.
- Implemented MUSIC-based Direction-of-Arrival (DoA) estimation and RSSI-based signal-strength analysis to compute azimuth angle, elevation angle, bearing direction, and approximate target distance.
- Used MATLAB and Simulink for signal processing, mathematical modeling, and simulation, and built a real-time monitoring dashboard with LED/buzzer-based threat alerting.
- Designed and trained an RF signal-classification model that ingests uploaded RF recordings and predicts signal category (Wi-Fi, Bluetooth, LTE, or Drone RF) with a confidence score, alongside live RSSI, peak power, and noise-floor readouts.

PROJECTS

Counter Drone RF Analyzer / Drone DF Simulator — GitHub

- Desktop application to upload and analyze RF recordings, visualize live power spectrum and spectrogram, and output AI-based drone-presence prediction with confidence, RSSI, peak power, and noise floor.

LilNest — Maternal Care Platform — Live Demo

- Built a full-stack maternal-health web application using React, Node.js, and MongoDB with authentication, responsive UI, and RESTful API integration between frontend and backend.

MotionBeatz — Gesture-Controlled Music Player

- Developed a computer-vision-based music player that uses hand-gesture recognition to control playback, volume, and track navigation.

PUBLICATIONS

- Co-author, *"NeuroCloak: Cognitive Digital Twin (CDT) for AI Systems,"* 2026 IEEE 8th International Conference on Cybernetics, Cognition and Machine Learning Applications (ICCCMLA) — currently under review.
- Proposed a Cognitive Digital Twin framework combining neuro-symbolic AI and metacognitive monitoring to make AI decision-making self-evaluating, bias-auditable, and transparent for high-stakes domains such as healthcare and defense.

OPEN SOURCE CONTRIBUTIONS

- **freeCodeCamp —** Contributed a verified, merged commit adding interactive examples to the functional-programming curriculum lesson (PR #63872).
- Active contributor to community-driven open-source repositories on GitHub, focused on improving real-world code quality and documentation.

CERTIFICATIONS

- MongoDB Associate Developer — Verified Certification
- NVIDIA — Certificate of Competency, "Accelerating End-to-End Data Science Workflows"

ACHIEVEMENTS

- Selected participant, IIT Madras Workshop
- Honors Roll Student — recognized for sustained academic excellence

TECHNICAL SKILLS

- **Languages:** Java, C++, Python, JavaScript, C
- **Frameworks:** React.js, Node.js, Express.js, FastAPI, Tailwind CSS
- **Databases:** MongoDB, PostgreSQL, MySQL
- **Developer Tools:** Git, GitHub, VS Code, Postman, Render
- **AI / ML:** Data Preprocessing & Feature Engineering, Computer Vision, Machine Learning Algorithms
- **Core CS:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
- **Soft Skills:** Problem-Solving, Team Collaboration, Communication, Project Management, Critical Thinking, Adaptability